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Emergent Higgs Field and the Schwarzschild Black Hole

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Abstract

The derivations presented in this paper suggest an intimate relationship between geometry and the electroweak sector at the Planck scale. A Lorentz-invariant maximally symmetric stochastically perturbed spacetime transformed to spherical coordinates reveals an emergent Schwarzschild metric, entirely a statistical structure of stochastic spacetime. Similarly, the transition from a maximally symmetric universe with a complex $SU(2)$ scalar doublet ϕ, comprising four independent real scalar fields with a zero vacuum expectation value (VEV), to spherical coordinates at the Planck scale reveals the spontaneously broken electroweak (EW) sector. Working in the unitarity gauge, the resulting EW potential can be simultaneously mapped in space at the Planck scale and across the EW sector. In space, the resulting EW potential includes a deep well within the Schwarzschild sphere and a shallow well just outside corresponding to an accretion disk. The same potential mapped in the EW space provides an entire family of possible sombrero hat potentials with fourth-order coupling specific to a point in space. At the minimum points of the potential in space, inside the Schwarzschild sphere and at the accretion disk, the λ corresponding to the Standard Model (SM) fourth-order coupling is instead derived as $\frac{\lambda}{5}$. The factor of $\frac{1}{5}$ is a simple consequence of the conservation of the EW VEV and the fact that the SM formulation of the EW potential does not account for situations where the perturbations in ϕ dominate. A more general formulation of the EW potential restores the SM quartic coupling and preserves λ in space. An emergent Higgs field inside the Schwarzschild black hole is found to directly relate to the stochastic spacetime fields normalized by the Schwarzschild radius. The corresponding Higgs vacuum has both a ground and excited state and the possibility of both positive and negative vacuum entropy. Finally, the scalar-field VEV degeneracy in EW space of the metastable Higgs vacuum appears instead differentiated in space with possible probability, tunneling, and entropy implications.

Keywords: emergent Higgs; Schwarzschild black hole; Higgs vacuum; vacuum entropy; spontaneous symmetry breaking; sombrero hat potential; stochastic spacetime; Smoluchovski equation; Schwarzschild horizon; Schwarzschild accretion disk

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Academic Editor: Maxim Yu. Khlopov

Received: 22 January 2026
Revised: 30 March 2026
Accepted: 1 April 2026
Published: 3 April 2026

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1. Introduction

The quest to unify Einstein's classical theory of general relativity with contemporary quantum field theories has motivated many new innovative ideas in cosmology and quantum field theory [1–5]. The challenge of bridging the vastly different scales between cosmology and particle physics is so fundamental that it inspires new approaches similarly considering fundamental changes or additions to existing theories. One area of such innovations involves incorporating the effects of randomness, or stochasticity. While many new models incorporate randomness into the spacetime manifold by perturbing the metric of spacetime, some new ideas consider perturbing spacetime itself [6–10].

The universal law of nature spanning both sides of the scale spectrum is the preservation of the speed of light, $U^2 = -1$. Recent research exploring invariance of the proper time functional to transformations with stochastic elements [11] has led to some interesting results in cosmology, including the possibility of an infinitely old universe and a resolution to the Hubble tension [12,13]. This paper continues this exploration to find an emergent Higgs field, a relationship between the Higgs field and the Schwarzschild radius, the possibility of a positive and negative Higgs vacuum entropy, and a novel look at the Higgs field renormalization and fermion mass.

We begin by placing an electroweak (EW) $SU(2)$ scalar field, ϕ, with basic quadratic scalar potential $m_\varphi^2 \phi^\dagger \phi$, corresponding to a zero EW vacuum expectation value (VEV), into a spacetime with stochastic fields. We find that the quadratic potential has a preferred direction parallel to the fields of the stochastic spacetime. Transforming to radial coordinates at the Planck scale further reveals a steep well potential inside the Schwarzschild radius, r_0, with a shallow potential well just outside.

The resulting derivations find an emergent Higgs field at the minimum of the steep well potential, a function of the ratio of the stochastic spacetime field and the Schwarzschild radius, and an emergent sombrero ϕ^4 potential with ϕ in the unitarity gauge. The potential is identical to the Standard Model (SM) Higgs potential, but with the SM $(\phi^\dagger \phi)^2$ coefficient λ instead derived with an accompanying factor of $\frac{1}{5}$, a simple consequence of the conservation of EW VEV and the fact that the SM formulation of the EW potential does not account for situations where the perturbations in ϕ dominate. A more general formulation of the EW potential restores the SM quartic coupling and preserves SM λ in space.

The transition from the maximally symmetric vacuum to the Higgs vacuum of the broken EW sector is only possible at a transition radius inside the Schwarzschild sphere. The drop from the excited state of the maximally symmetric vacuum with stochastic fields but with $\langle \phi \rangle = 0$ down to the ground state potential for the $SU(2)_L$ scalar field of the broken EW sector can occur naturally at this point. Consequently, EW potential can be naturally followed further down to its minimum value inside the Schwarzschild black hole and the deep well potential, or by tunneling to the second minimum of the potential at the accretion radius.

The two degenerate scalar field solutions with positive and negative VEV values ($\pm v$) in the EW sector map to the two minimum potential points in space, one corresponding to the minimum inside the Schwarzschild sphere and the other to the accretion area just outside the Schwarzschild sphere. In the EW sector, we cannot differentiate between these two fields. However, in space, they are distinct, both in terms of the character of their respective potential wells as well as in terms of their corresponding entropy and statistical probability.

The steep potential well inside the Schwarzschild sphere allows for the assumption of a highly dampened state and the application of the Smoluchovski diffusion equation [14–18]. Focusing on the region just inside the Schwarzschild horizon and with the help of path integral formalism, we derive the density of the Higgs fields in the vacuum to find it a function of $\frac{\mu^2 v^2}{k_B T}$, where μ and v are the traditional SM Higgs Lagrangian $\phi^\dagger \phi$ coefficient and ϕ expectation value parameters, respectively, and k_B, T are the traditional Boltzmann constant and temperature. We find two possible entropy vacuum states, corresponding to a positive and negative entropy along with the Green's function for the Higgs fields. The invariance of EW VEV requires a mixed state of the two $SU(2)_L$ Higgs fields and provides us with an estimate of relative probabilities for the two states corresponding to positive and negative entropy.

Finally, a brief consideration of the Dirac equation placed inside the stochastic spacetime provides an emergent coupling of fermions to the kinetic Higgs ∂h rather than the Higgs h, and allows for a gauge-invariant fermion mass term.

This paper is organized as follows. We first review the framework of stochastic spacetime perturbations, followed by the derivations of the emergent Higgs and the Schwarzschild black hole. We then consider the Higgs vacuum as an over-damped system in equilibrium with the help of path integral formalism and the Smoluchovski equation. We end with a brief discussion of applications and a summary.

Unless stated otherwise, we work in natural units, assuming a mostly positive Minkowski metric, and in local spatial coordinates at the Planck scale, $x^\mu = (t, \delta x^i)$ with $\delta x \sim l_P$. We assume that the vacuum is a system in equilibrium, approximating away time dependence in the scalar field and its potential. We work in a torsionless universe where the stochasticity in time is much smaller than in space and can be ignored [12,13]. Finally, we work under Itô calculus.

2. Framework of Stochastic Spacetime Perturbations

Invariance of the proper time functional enforces the fundamental law of nature, $U^2 = -1$, regardless of the frame of reference or the scale of the problem. Einstein's theory of general relativity (GR) is predicated on this requirement. Expanding the range of possible transformations to also include stochastic elements results in an expansion of the Christoffel connection, Friedmann equations, and Einstein field equations, where diffusion resulting from stochastic elements in spacetime together with geometry plays an important role in the evolution of the universe. A full implementation of this framework must also consider the impact perturbations in spacetime have on the momentum space.

The study of hydrodynamics and statistical mechanics often includes an element of uncertainty [19–34]. Commonly, a relativistic particle's four-momentum, p^μ, is allowed to exhibit some noise, explicitly modeled via a Langevin equation. The particle's location in spacetime, x^μ, is determined by p^μ, and thus must also carry some element of uncertainty. However, it is generally effectively treated as deterministic. This is mostly an acceptable approximation as the corresponding scale of uncertainty in spacetime tends to be far smaller than the scale of uncertainty in four-momentum. Nonetheless, it remains an important approximation that ignores any higher-order terms in the stochastic variables and is generally taken for granted.

A classical particle exhibiting Brownian motion can be quantified via a Langevin equation, a function of the particle's position in space $\mathbf{x}$ at some time t with some Gaussian noise $\mathbf{z}(t, \mathbf{x})$. The resulting momentum obtains a stochastic element as a derivative of $\mathbf{z}(t, \mathbf{x})$. In this case, it is typically assumed that both the position and the momentum at a particular point in space and time are entirely defined by the single stochastic variable $\mathbf{z}(t, \mathbf{x})$, effectively ignoring the definition of a derivative as an asymptotic limit subject to two different points in space and time and thus a function of potentially two different and possibly independent stochastic variables. Such a simplified approach is generally acceptable when considering a system originally in equilibrium perturbed by an event at a single point in space and time, but becomes problematic when there is an effective minimum length at the Planck scale where spacetime becomes unphysical. In this case, the derivative itself becomes a stochastic variable with an imperfect correlation to the perturbations in spacetime.

Alternative to such traditional applications of random factors directly modeled in spacetime or four-momentum but not directly in both, we consider recognizing and accounting for the dynamic effects of stochastic behavior in both. If the deterministic spacetime gains a Gaussian noise term, the resulting four-momentum must also be stochastic, with

none or some less-than-perfect correlation to the spacetime. This is the focus of the Robertson Walker minimum length (RWML) theory, with applications in stochastic calculus and cosmology [11–13].

We proceed with a review of the RWML theory relevant to the derivations regarding the Higgs field and EW potential.

2.1. Stochastic Spacetime Fields

Consider the possibility that each point in spacetime, x, also carries some very small Planck-scale uncertainty generated by a stochastic field unique to x, $\delta\xi$. We treat such small fluctuations as fundamental to the spacetime and would like to understand if and how stochastic fields $\delta\xi$ relate to the Higgs field. We will refer to such perturbations as either spacetime fields or Planck fields throughout this paper.

While many of the details of uncertainty in spacetime are explained in the literature [11–13], here, we introduce the concepts relevant to their application in a Higgs vacuum. We allow spacetime to take on an element of uncertainty $\delta\xi$ over some segment in spacetime δx, which we assume to be at the Planck scale:

$$\delta\tilde{x} \equiv \delta x + \delta\xi \tag{1}$$

$$\delta\xi \sim \mathcal{N}(0, \sigma_\xi\sqrt{\delta t}), \quad \langle\delta\xi^\mu\rangle_{\delta\xi} = 0 \tag{2}$$

where δt scales with Planck length, and σ_ξ is the Planck field's diffusion parameter [11], here chosen to represent the length of the vector Planck field and thus following a slightly different nomenclature than the previous papers on stochastic spacetime. Also note that the expectation values are taken with respect to the Planck fields, $\delta\xi$. The choice of this notation is to avoid confusion with the expectation values relative to the kinetic stochastic fields, $\partial\delta\xi$.

Most generally, we can quantize the uncertainty in spacetime via the following two point functions that we separate out across space and time:

$$\langle\delta\xi^0\delta\xi^0\rangle_{\delta\xi} = -\sum_i \sigma_\epsilon^2\delta x^i, \quad \langle\delta\xi^{i\,2}\rangle_{\delta\xi} = \sigma_\xi^2\delta t \tag{3}$$

Here, the parameters σ_ϵ and σ_ξ (both of dimension $L^{\frac{1}{2}}$) are measures of the magnitude of perturbations in time and space, respectively. The relationships in Equation (3) can be simplified further under the assumption of a torsionless universe requiring $\sigma_\epsilon^2 \approx 0$ [13], giving us $\delta\xi^\mu \approx (0, \delta\xi^i)$ and $\langle\delta\xi^2\rangle_{\delta\xi} = \sigma_\xi^2\delta t$, consistent with the magnitude of the perturbations provided by Equation (2). (Note that the notation here caters to topics covered in this paper and differs slightly from the normalizations used in [11–13].)

The magnitude of each spatial uncertainty in a maximally symmetric universe is one third of the total vector length, $\langle\delta\xi^i\delta\xi^j\rangle_{\delta\xi} = \frac{1}{3}g^{ij}\sigma_\xi^2\delta t$. Lorentz invariance of a stochastic spacetime [11] allows us to continuously rotate into a different frame, or alternatively, perform a coordinate transformation resulting in the three spatial components exhibiting non-uniform magnitudes of diffusion. Regardless of the choice of frame, however, the vector length must always be preserved, $\langle\delta\xi^2\rangle_{\delta\xi} = \sigma_\xi^2\delta t$.

Taking partial derivatives of the stochastic fields gives us the kinetic stochastic fields, $\partial\delta\xi$, with $\langle\partial_\mu\delta\xi^\nu\rangle_{\partial\delta\xi} = 0$. We can similarly simplify the two-point functions for the kinetic stochastic terms by defining

$$\langle\partial_\alpha\delta\xi^0\partial_\beta\delta\xi^0\rangle_{\partial\delta\xi} \approx 0, \quad \langle\partial_\alpha\delta\xi^i\partial_\beta\delta\xi^j\rangle_{\partial\delta\xi} = 2\delta_{\alpha\beta}g^{ij}c_\xi\sigma_\xi^2\delta t \tag{4}$$

where now, we also have a diffusive parameter corresponding to the kinetic stochastic fields, $c_{\tilde{\zeta}} \sim \frac{1}{3\delta t^2}$, of dimension L^{-2}, and where the expectation values are taken relative to the kinetic Planck fields.

Most generally, the kinetic stochastic fields and the spacetime fields will have some correlation [11–13]. We leave the effects of nonzero correlation to future research. In this paper, for simplicity and without loss of generality, we set this correlation to approximately zero for all i, j as follows:

$$\langle \delta\tilde{\zeta}^i \partial_\mu \delta\tilde{\zeta}^j \rangle_{\delta\tilde{\zeta},\, \partial\delta\tilde{\zeta}} \approx 0 \tag{5}$$

Note that assuming a zero correlation allows us to separate the expectation values with regards to the Planck fields and their kinetic counterparts, as follows:

$$\langle \delta\tilde{\zeta}^i \rangle_{\partial\delta\tilde{\zeta}} = \delta\tilde{\zeta}^i, \quad \langle \partial_\mu \delta\tilde{\zeta}^i \rangle_{\delta\tilde{\zeta}} = \partial_\mu \delta\tilde{\zeta}^i \tag{6}$$

The traditional approach to kinetic stochastic fields at x assumes that they are entirely defined by the distribution of the stochastic fields at x: $\partial_\mu \delta\tilde{\zeta}^\alpha_x = \lim_{\delta x^\mu \to 0}(\delta\tilde{\zeta}^\alpha_{x+\frac{\delta x^\mu}{2}} - \delta\tilde{\zeta}^\alpha_{x-\frac{\delta x^\mu}{2}}) = f(\delta\tilde{\zeta}^\alpha_x)$, where f is some function. However, this is a problematic approach when the stochastic spacetime introduces an effective minimum length at the Planck scale. Taking the limit for $\delta x^\mu \to 0$ is no longer feasible given the unphysical character of spacetime below some minimum value. In this case, the derivatives of stochastic variables are explicitly defined by two points in spacetime and cannot have a perfect correlation to uncertainty in spacetime, $\partial_\mu \delta\tilde{\zeta}^\alpha_x \neq f(\delta\tilde{\zeta}^\alpha_x)$.

Here, we deviate from this approach in recognizing that $\partial\delta\tilde{\zeta}$ are most generally also stochastic fields—albeit of different dimension to $\delta\tilde{\zeta}$—with some correlation to $\delta\tilde{\zeta}$. Carrying this out allows us to correctly treat their two point functions in a universe with an effective minimum length, as well as account for how both might simultaneously play a role in the evolution of a system.

2.2. Schwarzschild Metric as an Emergent Statistical Geometry

Spacetime with Gaussian perturbations in Cartesian coordinates is maximally symmetric, Lorentz invariant, [11] and consistent with a vacuum that is homogeneous and isotropic. In Cartesian coordinates, Planck fields have a zero VEV, $\langle \delta\tilde{\zeta}^\mu \rangle = 0$. The transition to spherical coordinates has the effect of folding spacetime onto itself, as follows: $r = \sqrt{x^{i\,2}}$. Spacetime perturbations are reorganized to reveal nonzero perturbation correlations in the neighborhood of $r \approx 0$—otherwise not visible in Cartesian coordinates—and result in a nonzero VEV in radial perturbations.

An effective Schwarzschild geometry emerges, the result of a statistical structure in the proper-time functional and not the result of curvature from the Einstein equations. A source term defined entirely by the magnitude of spacetime perturbations σ^2 is revealed but the diffeomorphism invariance is preserved.

The transition to spherical coordinates gives the radial component a nonzero expectation value and an effective singularity at $r = 0$ [11,13]:

$$\delta\tilde{x}^i = \delta x^i + \delta\tilde{\zeta}^i \quad \longrightarrow \quad \delta\tilde{r} = \delta r + \frac{\sigma^2_{\tilde{\zeta}} \delta t}{2r} + \delta\tilde{\zeta}^r \tag{7}$$

The full derivation of the resulting emergent Schwarzschild metric, physically possible only going forward in time, is provided in [13]. Here, we summarize only the results necessary to formulate and interpret the scalar potential placed in stochastic spacetime.

Given a forward-looking arrow of time, the Schwarzschild metric emerges as [13]

$$\frac{dt}{dr} = \frac{1}{1 - \frac{\sigma_\xi^2}{2r}} \tag{8}$$

In [13], the transition to spherical coordinates results in the derivation of Equation (8), where σ_ξ can be identified as $4G\delta m$, the mass of the Schwarzschild metric. Additional derivation of the curvature parameter resulting from a coupling of the Hubble parameter and diffusion induced by stochastic spacetime [13] further defines the scale of $\sigma_\xi\ l_p$ [13]. The resulting $\delta m \sim \frac{1}{4\sqrt{G}}$ scales with the Planck mass. Here, we use the former relationship $\sigma_\xi\ l_p$ to identify the Schwarzschild radius at the Planck scale, $r_0 = \frac{\sigma_\xi^2}{2}$. With $\sigma_\xi^2 \sim \delta t$ [13], we can also write $r_0^2 = \frac{\sigma_\xi^2 \delta t}{4}$. These relationships will prove useful.

2.3. The Importance of Ensuring Consistent Order of δt

We first consider the EW scalar field doublet ϕ prior to spontaneous symmetry breaking. The EW scalar can be expressed in terms of four real and independent scalar fields. We focus on one of these real scalar fields, $\varphi(\tilde{x})$, and consider how it transforms as we transition from one reference frame to another.

We place φ into a stochastic spacetime $\tilde{x}$ and begin by expanding $\varphi(\tilde{x})$ in its power of the stochastic spacetime fields, $\delta\xi$, to order δt. A corresponding quadratic scalar potential is thus set to order δt^2. We impose a zero expectation value on the scalar field prior to spontaneous symmetry breaking, and find that it results in the expected scalar field potential at its minimum along the direction of the Planck fields. We assume zero force in the direction of the Planck fields—this must be upheld pre-spontaneous symmetry breaking—and this must hold at all times.

As we consider how the potential transforms under a change in the reference frame and in formulating the Fokker–Planck equation for the Higgs vacuum density, we are required to derive various derivatives of the scalar field and the potential. In accomplishing this, we must understand the appropriate order of δt for the derivatives as well as for expectation values of such derivatives.

If we choose to work with a scalar field with granularity of order n, δt^n, we must ensure that throughout our analysis we do not incorporate higher orders of δt than appropriate given the order of δt chosen for the scalar field. Including terms of higher order in δt results in introducing errors rather than achieving greater precision.

Consider a scalar field in the stochastic spacetime, $\varphi(\tilde{x})$, where we choose to expand the scalar field to the n-th order in δt, δt^n. If we are to formulate any partial derivatives of this scalar field, for example, $\partial_i \varphi(\tilde{x})$ or $\partial^2 \varphi(\tilde{x})$, we must understand the maximum appropriate granularity of such terms. Specifically, as a partial derivative ∂_μ is, by definition, of order $\frac{1}{\delta t}$, we have

$$\varphi(\tilde{x}) \sim \delta t^n \quad \Rightarrow \quad \partial^m \varphi(\tilde{x}) \sim \delta t^{n-m} \tag{9}$$

In other words, if the scalar field has a granularity corresponding to δt^n, then an $m - th$ partial derivative will have a much less sensitive granularity of order δt^{n-m}.

Proper treatment of the order of δt is crucial if we are to work with a system requiring a zero net force in some direction, for example $\partial_i \langle U(\tilde{x}) \rangle_{\delta\xi} = 0$. From Equation (9), we see that regardless of the δt granularity in the potential, this boundary condition on the potential will be one order lower in δt relative to the potential. As such, the boundary condition can be characterized as 'loose', possibly leading to an entire family of solutions for the potential.

3. Emergent Higgs and the Schwarzschild Black Hole

Prior to spontaneous symmetry breaking the complex SU(2) scalar doublet of the EW sector, ϕ, is given by

$$\phi = \frac{1}{\sqrt{2}}\begin{pmatrix}\phi^+\\ \phi^0\end{pmatrix} \quad = \frac{1}{\sqrt{2}}\begin{pmatrix}\varphi_3 + i\varphi_4\\ \varphi + i\varphi_1\end{pmatrix}, \tag{10}$$

where $\varphi, \varphi_i (i = 1, 2, 3)$ are independent real scalar fields. Each of these four real fields must have a zero VEV in the maximally symmetric universe prior to EW spontaneous symmetry breaking, enforced by the simple quadratic scalar potential. As these four fields are independent, we can isolate each of the fields and subject them to their own quadratic potential:

$$U = m_\varphi^2 \phi^\dagger \phi = \frac{1}{2}(U_0 + U_1 + U_2 + U_3), \quad U_0 \equiv m_\varphi^2 \varphi^2, U_i \equiv m_\varphi^2 \varphi_i^2 \tag{11}$$

With the expectation of spontaneous symmetry breaking allowing us to rotate to the unitarity gauge, we focus our attention on one of the four independent fields, φ.

Prior to spontaneous symmetry breaking, a real scalar field of order δt placed in a vacuum with stochastic spacetime, $\varphi(\tilde{x})$, and in a quadratic potential, $U_0(\tilde{x}) = m_\varphi \varphi(\tilde{x})^2$, must have a zero VEV. We find that requiring a zero expectation value for the scalar field results in $\partial_{\hat{n}}^2 \varphi(x) = -\frac{2\varphi(x)}{\sigma_{\tilde{\xi}}^2 \delta t}$, where $\hat{n}$ is the direction of the Planck field $\delta\tilde{\xi}$. Consequently, we also find that $\delta\tilde{\xi} \cdot \partial\langle U_0(\tilde{x})\rangle_{\delta\tilde{\xi}} = 0$. The expected potential has a preferred direction: it is minimized in the direction of the Planck field.

In the transformation from Cartesian to spherical coordinates, $\delta\tilde{\xi}^i \to \frac{\sigma^2 \delta t}{2r} + \delta\tilde{\xi}^r$, the scalar field acquires a nonzero VEV. The coordinate transition reveals a spontaneously broken EW symmetry. We express the scalar field as a function of $\tilde{r}$ in place of $\tilde{x}$ and the potential transforms to some new function, now in spherical coordinates $U(\tilde{r})$.

At some point in space $\tilde{r}_T$ ($\tilde{x}_T$ in Cartesian coordinates), both $U_0(\tilde{x})$ and the new potential $U(\tilde{r})$ must 'coexist' and must both be at their minimum value relative to the direction of $\delta\tilde{\xi}$. We thus require $\partial_{\hat{n}}^2 \varphi(x) = -\frac{2\varphi(x)}{\sigma_{\tilde{\xi}}^2 \delta t}$, ensuring $\delta\tilde{\xi} \cdot \partial\langle U_0(\tilde{x})\rangle_{\delta\tilde{\xi}} = 0$ still holds. At $\tilde{r} = \tilde{r}_T$, we must also require $\delta\tilde{\xi} \cdot \partial\langle U(\tilde{r})\rangle_{\delta\tilde{\xi}} = 0$. We can summarize these conditions as follows:

$$\langle\varphi(\tilde{x}_T)\rangle_{\delta\tilde{\xi}} = 0 \quad \Rightarrow \quad \partial_{\hat{n}}^2 \varphi(x_T) = -\frac{2\varphi(x_T)}{\sigma_{\tilde{\xi}}^2 \delta t} \quad \Rightarrow \quad \delta\tilde{\xi} \cdot \partial\langle U_0(\tilde{x}_T)\rangle_{\delta\tilde{\xi}} = 0 \tag{12}$$

$$\delta\tilde{\xi} \cdot \partial\langle U(\tilde{r}_T)\rangle_{\delta\tilde{\xi}} = 0 \tag{13}$$

As the scalar field is set to order δt, the quadratic potential is set to order δt^2, requiring the formulation of $\partial\langle U(\tilde{r}_T)\rangle_{\delta\tilde{\xi}}$ in Equation (12) to order δt.

After the transition at r_T, we can remove the requirements imposed by Equation (12) on $\varphi(\tilde{x})$ and $U_0(\tilde{x})$, and we then only require the minimum of the potential in the $\delta\tilde{\xi}$ direction imposed on $U(\tilde{r})$, as shown in Equation (13).

A brief summary of the boundary conditions throughout the transition is shown in Figure 1.

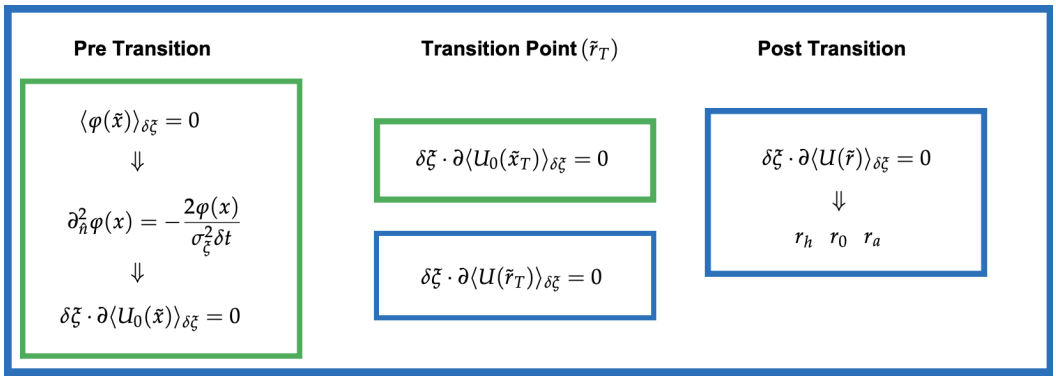

Figure 1. High-level view of the transition boundary conditions. Boundary conditions pre-transition are shown inside green boxes, and post transition in blue.

We now follow with the details of the derivations and results.

3.1. Scalar Field and Vacuum Potential Prior to Transition

To formulate a real scalar field and its potential in a stochastic Cartesian spacetime prior to the coordinate transition we expand $\varphi(\tilde{x})$ in terms of its deterministic and stochastic components to order δt [13] and take the following expectation value:

$$\varphi(\tilde{x}) = \varphi(x) + \partial_\mu\varphi(x)\ \delta\tilde{\xi}^\mu + \frac{1}{2}\partial_\mu\partial_\nu\varphi(x)\ \delta\tilde{\xi}^\mu\delta\tilde{\xi}^\nu \tag{14}$$

$$\langle\varphi(\tilde{x})\rangle_{\delta\tilde{\xi}} = \varphi(x) + \frac{1}{2}\partial^2\varphi(x)\ \frac{1}{3}\sigma^2_{\tilde{\xi}}\delta t \tag{15}$$

Alternatively, we can simplify the formulation by choosing a local frame continuously rotated such that $\delta\tilde{\xi}$ always points in the **z**-direction, $\delta\tilde{\xi}^\mu \approx (0, 0, 0, \delta\tilde{\xi}^{\hat{z}})$. We can repeat Equation (14) in this frame to obtain

$$\varphi(\tilde{x}) = \varphi(x) + \partial_{\hat{z}}\varphi(x)\ \delta\tilde{\xi}^{\hat{z}} + \frac{1}{2}\partial^2_{\hat{z}}\varphi(x)\ \delta\tilde{\xi}^{\hat{z}\ 2} \tag{16}$$

Regardless of the choice of frame, the length of $\delta\tilde{\xi}$ must be preserved, $\langle\delta\tilde{\xi}^2\rangle_{\delta\tilde{\xi}} = \sigma^2_{\tilde{\xi}}\delta t$, and thus, in the choice frame, we have $\langle\delta\tilde{\xi}^{\hat{z}\ 2}\rangle_{\delta\tilde{\xi}} = \sigma^2_{\tilde{\xi}}\delta t$. This gives us the second equation

$$\langle\varphi(\tilde{x})\rangle_{\delta\tilde{\xi}} = \varphi(x) + \frac{1}{2}\partial^2_{\hat{z}}\varphi(x)\ \sigma^2_{\tilde{\xi}}\delta t \tag{17}$$

In a maximally symmetric vacuum, the scalar field must look the same in any direction. In this case, $\partial^2_{\hat{x}}\varphi(x) = \partial^2_{\hat{y}}\varphi(x) = \partial^2_{\hat{z}}\varphi(x) = \frac{1}{3}\partial^2\varphi(x)$, and we see that Equations (15) and (17) are in fact identical.

We can thus generalize to any choice frame with $\delta\tilde{\xi}$ in some direction $\hat{n}$, $\delta\tilde{\xi} = \delta\tilde{\xi}^{\hat{n}}$, as follows:

$$\delta\tilde{\xi} = \delta\tilde{\xi}^{\hat{n}} \quad \Rightarrow \quad \varphi(\tilde{x}) = \varphi(x) + \partial_{\hat{n}}\varphi(x)\ \delta\tilde{\xi}^{\hat{n}} + \frac{1}{2}\partial^2_{\hat{n}}\varphi(x)\ \delta\tilde{\xi}^2 \tag{18}$$

$$\langle\varphi(\tilde{x})\rangle_{\delta\tilde{\xi}} = \varphi(x) + \frac{1}{2}\partial^2_{\hat{n}}\varphi(x)\ \sigma^2_{\tilde{\xi}}\delta t \tag{19}$$

Requiring a zero expectation value for the scalar field prior to the EW spontaneous symmetry breaking, we obtain

$$\langle\varphi(\tilde{x})\rangle_{\delta\tilde{\xi}} = 0 \quad \Rightarrow \quad \partial^2_{\hat{n}}\varphi(x) = -\frac{2\varphi(x)}{\sigma^2_{\tilde{\xi}}\delta t} \tag{20}$$

Finally, the scalar field with a zero expectation value is given by

$$\varphi(\tilde{x}) = \varphi(x)(1 - \frac{\delta\xi^{\hat{n}\,2}}{\sigma_{\tilde{\xi}}^2 \delta t}) + \partial_{\hat{n}}\varphi(x)\ \delta\xi^{\hat{n}} \tag{21}$$

The corresponding quadratic scalar field potential and its expectation value then follow as

$$U_0(\tilde{x}) \equiv m_\varphi^2 \varphi(\tilde{x})^2 \tag{22}$$

$$\langle U_0(\tilde{x})\rangle_{\delta\tilde{\xi}} = 2m_\varphi^2\{\varphi(x)^2 + \frac{1}{2}(\partial_{\hat{n}}\varphi(x))^2\sigma_{\tilde{\xi}}^2\delta t\} \tag{23}$$

Note that the potential is minimized in the direction of $\delta\tilde{\xi}$, as follows:

$$\partial_{\hat{n}}\langle U_0(\tilde{x})\rangle_{\delta\tilde{\xi}} = 0 \quad \Rightarrow \quad \delta\tilde{\xi}\cdot\partial\langle U_0(\tilde{x})\rangle_{\delta\tilde{\xi}} = 0 \tag{24}$$

We have chosen a particular frame for $\delta\tilde{\xi}$. However, the algebra of stochastic spacetime is Lorentz invariant [11], allowing us to rotate to any frame of choice. The relationship in Equation (24) must remain true regardless of this choice: the potential must be at its minimum in the direction of the Planck fields.

3.2. Spontaneously Broken EW Symmetry Revealed at the Point of Transition

The transition from Cartesian to spherical coordinates, $\delta\xi^{\hat{n}} \to \delta\xi^r + \frac{\sigma^2\delta t}{2r}$, reveals the scalar field with a nonzero VEV given by

$$\varphi(\tilde{r}) = \varphi\ (1 - \frac{\delta\xi^{r\,2}}{\sigma_{\tilde{\xi}}^2\delta t}) + \delta\xi^r\ (\partial_r\varphi - \frac{\varphi}{r}) + \frac{\sigma_{\tilde{\xi}}^2\delta t}{2r}\ (\partial_r\varphi - \frac{\varphi}{2r}) \tag{25}$$

where the scalar field is now expressed as a function in spherical coordinates, and where we simplified the notation to $\varphi \equiv \varphi(r)$, with $\delta\xi^\mu = (0, \delta\xi^r, 0, 0)$.

The corresponding potential now becomes

$$\begin{aligned} U(\tilde{r}) &= m_\varphi^2\varphi(\tilde{r})^2 \\ &= m_\varphi^2\{\varphi^2\ (1 - \frac{\delta\xi^{r\,2}}{\sigma_{\tilde{\xi}}^2\delta t})^2 + 2\varphi\ \delta\xi^r(\partial_r\varphi - \frac{\varphi}{r})(1 - \frac{\delta\xi^{r\,2}}{\sigma_{\tilde{\xi}}^2\delta t}) + \frac{\sigma_{\tilde{\xi}}^2\delta t}{r}\varphi(\partial_r\varphi - \frac{\varphi}{2r})(1 - \frac{\delta\xi^{r\,2}}{\sigma_{\tilde{\xi}}^2\delta t}) \\ &+ \delta\xi^{r\,2}(\partial_r\varphi - \frac{\varphi}{r})^2 + \frac{\sigma_{\tilde{\xi}}^2\delta t}{r}\delta\xi^r(\partial_r\varphi - \frac{\varphi}{r})(\partial_r\varphi - \frac{\varphi}{2r}) + \frac{\sigma_{\tilde{\xi}}^4\delta t^2}{4r^2}(\partial_r\varphi - \frac{\varphi}{2r})^2\} \end{aligned} \tag{26}$$

At the transition point r_T (x_T) both $\langle U_0(\tilde{x})\rangle_{\delta\tilde{\xi}}$ and $\langle U(\tilde{r})\rangle_{\delta\tilde{\xi}}$ must be at their minimum values in the direction of $\tilde{\xi}$, expressed in Cartesian and spherical coordinates, respectively. The potential prior to transition, $U_0(\tilde{x})$, is minimized at this point as long as $\partial_{\hat{n}}^2\varphi = -\frac{2\varphi}{\sigma_{\tilde{\xi}}^2\delta t}$. To be able to use this information and apply it to $U(\tilde{r})$ at r_T, we must first formulate $\partial_r^2\varphi$ in terms of $\partial_{\hat{r}}^2\varphi$. Once equipped with this information, we can formulate $\partial_r\langle U(\tilde{r})\rangle_{\delta\tilde{\xi}} = 0$ and solve for $\partial_r\varphi$ to finally obtain a solution for $\langle U(\tilde{r})\rangle_{\delta\tilde{\xi}}$ at the point of transition.

From Equation (26), we can easily read off the expectation value of the potential, as follows:

$$\langle U(\tilde{r})\rangle_{\delta\tilde{\xi}} = m_\varphi^2\{2\varphi^2 + \sigma_{\tilde{\xi}}^2\delta t\ (\partial_r\varphi - \frac{\varphi}{r})^2 + \frac{\sigma_{\tilde{\xi}}^4\delta t^2}{4r^2}(\partial_r\varphi - \frac{\varphi}{2r})^2\} \tag{27}$$

To formulate $\partial_r^2\varphi$, we use the Laplacian component in the radial direction together with $\partial_{\tilde{r}}^2\varphi = -\frac{2\varphi}{\sigma_{\tilde{\xi}}^2\delta t}$, which must hold at r_T as follows:

$$\partial_{\tilde{r}}^2\varphi = \frac{1}{r^2}\partial_r(r^2\partial_r\varphi) = \frac{2}{r}\partial_r\varphi + \partial_r^2\varphi \quad \Rightarrow \quad \partial_r^2\varphi = -\frac{2\varphi}{\sigma_{\tilde{\xi}}^2\delta t} - \frac{2}{r}\partial_r\varphi \tag{28}$$

$$\varphi \sim \delta t^1 \quad \Rightarrow \quad \partial_r^2\varphi \sim \delta t^{-1} \quad \Rightarrow \quad \partial_r^2\varphi = -\frac{2\varphi}{\sigma_{\tilde{\xi}}^2\delta t} + O(\delta t^0) \tag{29}$$

where we make sure to formulate the second-order partial derivative to the appropriate order of δt given that we defined the scalar function to order δt^1.

Equipped with $\partial_r^2\varphi = -\frac{2\varphi}{\sigma_{\tilde{\xi}}^2\delta t}$, we can minimize the expectation value of the potential in the radial direction as follows:

$$\partial_r\langle U(\tilde{r})\rangle_{\delta\tilde{\xi}} = m_\varphi^2\{-2\frac{\sigma^2\delta t}{r}\partial_r\varphi^2 + 3\frac{\sigma^2\delta t}{r^2}\varphi\partial_r\varphi + \frac{\varphi^2}{r}(4 - \frac{3}{2}\frac{\sigma^2\delta t}{r^2})\} + O(\delta t^2) \tag{30}$$

where we have again ensured a proper order of δt given that the potential is provided to order δt^2. At r_T, we must have $\partial_r\langle U(\tilde{r})\rangle_{\delta\tilde{\xi}^r} = 0$ and can thus solve for $\partial_r\varphi$ to obtain the following two possible solutions:

$$\partial_r\varphi_\pm = \frac{\varphi}{r}\{\frac{3}{4} \pm \sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}}\} \tag{31}$$

where $r_0 = \sqrt{\frac{\sigma_{\tilde{\xi}}^2\delta t}{4}}$ is the Schwarzschild radius [11]. Note that minimizing the expected potential puts a lower bound on the value for r: $r \geq \sqrt{\frac{3}{8}}r_0$, making the region inside the minimum radius sphere unphysical.

There are two possible solutions for the expected potential, corresponding to the two possible solutions for $\partial_r\varphi$:

$$\langle U(\tilde{r})_\pm\rangle_{\delta\tilde{\xi}^r} = 2m_\varphi^2\varphi^2\{1 + 2\frac{r_0^2}{r^2}(\frac{1}{4} \mp \sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}})^2 + 2\frac{r_0^4}{r^4}(\frac{1}{4} \pm \sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}})^2\} \tag{32}$$

The plot of the expected potential for the two possible states is shown in Figure 2, expressed as a ratio of the potential to $m_\varphi\varphi^2$, $\frac{\langle U_\pm(\tilde{r})\rangle_{\delta\tilde{\xi}}}{m_\varphi^2\varphi^2}$, plotted across the squared ratio of the radius to r_0, $\frac{r^2}{r_0^2}$. Inside the Schwarzschild sphere, $r < r_0$, the ground state is defined by U_-, while U_+ is the excited state. Outside the sphere, U_- is the excited state while U_+ is the ground state.

The resulting character of the two potential wells, one inside r_0 and the other outside at $r \approx 3.10r_0$, albeit much weaker, resembles a Schwarzschild black hole with the horizon at $r = r_0 = \frac{\sigma\sqrt{\delta t}}{2}$ and an accretion disk. The two solutions for the potential cross at two points: the first point corresponds to the minimum radius with a physical solution, $r^2 = \frac{3}{8}r_0^2$, and the second occurs at $r = r_0$.

The minimum of the steep well potential corresponding to U_- inside r_0 occurs at $r^2 = (5 - \sqrt{21})r_0^2$ ($r \approx 0.65r_0$), while the minimum of the weak potential outside the Schwarzschild radius occurs at $r^2 = (5 + \sqrt{21})r_0^2$ ($r \approx 3.10r_0$), corresponding to a black hole accretion disk, assuming a relatively flat scalar field in this regions. Note that a jump to the excited state inside the Schwarzchild sphere allows for an escape to the outside of the Schwarzchild black hole, with quantum tunneling providing another means of reaching the minimum at the accretion disk.

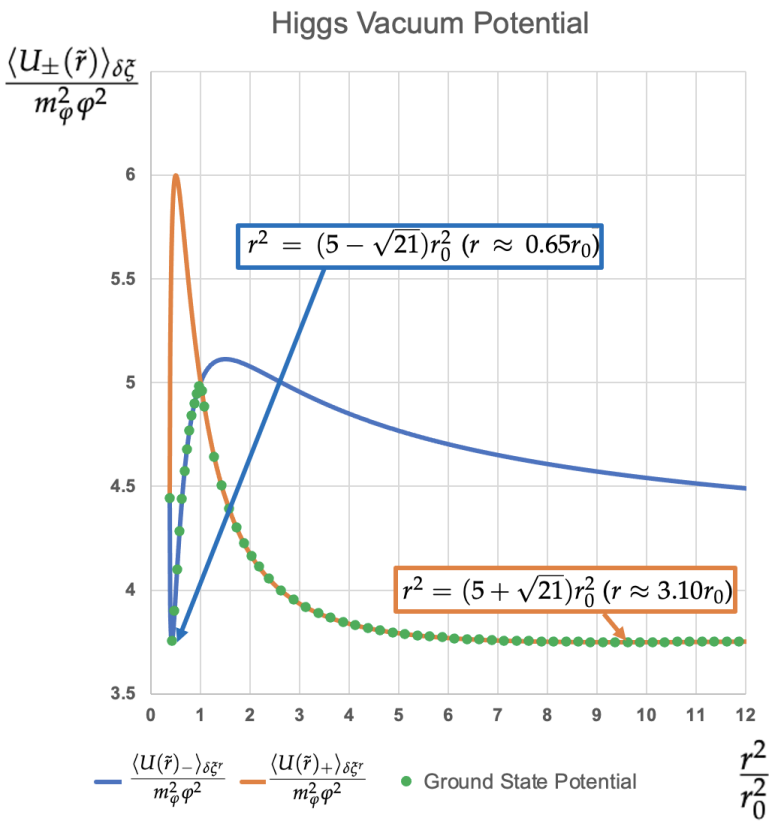

Figure 2. Potential post-transition corresponding to the ground and excited state with minimums at $r = \sqrt{5 \pm \sqrt{21}}$.

It is also helpful to write down the full formulation of the potential as follows:

$$\begin{aligned} U(\tilde{r})_\pm = m_\varphi^2\varphi^2\{1 &+ 4\frac{r_0^2}{r^2}\left(\frac{3}{4} \pm \sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}}\right) + 2\frac{r_0^4}{r^4}\left(-\frac{1}{4} \pm \sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}}\right) \\ &- \frac{\delta\xi^r}{r}\left(\frac{1}{4} \mp \sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}}\right)\left(2 + 4\frac{r_0^2}{r^2}\left(\frac{1}{4} \pm \sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}}\right)\right) \\ &- \frac{3}{2}\frac{\delta\xi^{r2}}{r^2}\left(\frac{1}{4} \pm \sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}}\right) + \frac{1}{2}\frac{\delta\xi^{r3}}{rr_0^2}\left(\frac{1}{4} \mp \sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}}\right) + \frac{1}{16}\frac{\delta\xi^{r4}}{r_0^4}\} \end{aligned} \tag{33}$$

Note that Equation (31) together with Equation (23) also provides two solutions for the potential corresponding to the maximally symmetric vacuum, $\langle U_0(\tilde{x})_\pm\rangle_{\delta\xi}$, but only in the region of possible transition.

3.3. The Schwarzschild Black Hole

Consider the ground state potential at its minimum value in the neighborhood of $r = r_h \equiv r_0\sqrt{5-\sqrt{21}}$. We substitute $\partial_r\varphi_-$ for $\partial_r\varphi$, and r_h for r in Equation (46), and take advantage of several helpful relationships at this minimum value to obtain

$$\frac{r_0^2}{r_h^2}\left(\frac{1}{4}-\sqrt{\frac{r_h^2}{2r_0^2}-\frac{3}{16}}\right)=\frac{1}{4} \tag{34}$$

$$\frac{r_0}{r_h}-\frac{r_h}{2r_0}=\sqrt{\frac{3}{2}} \tag{35}$$

$$\frac{r_0}{r_h}\left(\frac{1}{4}+\sqrt{\frac{r_h^2}{2r_0^2}-\frac{3}{16}}\right)=\frac{1}{2}\sqrt{\frac{3}{2}} \tag{36}$$

With the help of Equations (33)–(36), we obtain the ground state potential at its minimum value inside the steep well, perturbed by the stochastic field at the point in space $\delta\xi^r = \delta\xi^r_{r_h}$:

$$U(\tilde{r}) \to U(r_h, \delta\xi^r) = m_\varphi^2 \varphi_{r_h}^2 \{\frac{9}{4} - \frac{3}{2}\sqrt{\frac{3}{2}}\frac{\delta\xi^r}{r_0} - \frac{3}{8}\frac{\delta\xi^{r\,2}}{r_0^2} + \frac{1}{4}\sqrt{\frac{3}{2}}\frac{\delta\xi^{r\,3}}{r_0^3} + \frac{1}{16}\frac{\delta\xi^{r\,4}}{r_0^4}\} \tag{37}$$

Performing a variable transformation, $\tilde{z} \equiv \sqrt{\frac{3}{2}} + \frac{\delta\xi^r}{r_0} = \sqrt{\frac{3}{2}}(1+\sqrt{\frac{2}{3}}\frac{\delta\xi^r}{r_0})$ at r_h, we can write the potential as a function of $\tilde{z}$ as follows:

$$U(r_h, \tilde{z}) = const. + m_\varphi^2 \varphi_{r_h}^2 \{-\frac{15}{16}\tilde{z}^2 + \frac{1}{16}\tilde{z}^4\} \tag{38}$$

We can now promote the potential back to EW isospin algebra, with an SU(2) scalar doublet ϕ_s in the unitarity gauge defined by

$$\phi_s = \frac{1}{\sqrt{3}}\begin{pmatrix} 0 \\ v\tilde{z} \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v + h_s \end{pmatrix}, \quad h_s \equiv v\sqrt{\frac{2}{3}}\frac{\delta\xi^r}{r_0} \tag{39}$$

and an expectation value $\langle\phi_s\rangle_{\delta\xi^r} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v \end{pmatrix}$, but only if we can formulate $v\tilde{z}$ as a Lorentz-invariant scalar quantity.

The magnitude of the scalar field at the minimum of the potential, in this case the point r_h, is quantified by $v\tilde{z}$. Consider what this quantity represents. First, note that $\tilde{z}$ can be expressed as $\tilde{z} = \frac{1}{r_0}\delta\tilde{x}^r = \frac{1}{r_0}(\delta x_{r_h} + \delta\xi^r)$, where $\delta x_r \equiv \delta x(r)$. The field is highly constrained by the steep well around the minimum of the potential, allowing us to estimate the field strength with $\rho \sim \frac{\Delta\rho}{\Delta x}\Delta\tilde{x} \to \delta\tilde{x}^r\partial_r\rho$. With $v_r \equiv r_0\partial_r\rho \approx v$, we see that we can approximate ρ as a Lorentz-invariant scalar given by $\rho \approx \delta\tilde{x}^\mu\partial_\mu\rho = v_\mu\tilde{z}^\mu$ at the minimum of the steep well potential, with $v_\mu \equiv r_0\partial_\mu\rho$ and $\tilde{z}^\mu \equiv \frac{1}{r_0}\delta\tilde{x}^\mu$.

It is important to recognize that the magnitude of the scalar field ρ includes contributions both from h_s and v. SM assumes that v dominates over h_s, i.e., $v^2 >> \langle h_s^2\rangle_{\delta\xi}$, and $\rho_{vev} = \sqrt{\langle\rho^2\rangle_{\delta\xi}} = \sqrt{\langle v^2 + h_s^2\rangle_{\delta\xi}} \approx v$ is the positive VEV of the field.

Substituting in for $\tilde{z}^2$ in Equation (38), we obtain a sombrero hat potential identical to the Standard Model potential except for the factor of $\frac{1}{5}$ applied to the SM $\lambda \equiv \frac{\mu^2}{v^2}$:

$$U_-^{SU(2)}(r_h) = const. - \mu^2\phi_s^\dagger\phi_s + \frac{\lambda}{5}(\phi_s^\dagger\phi_s)^2, \quad \mu^2 v^2 \equiv \frac{45}{16}m_\varphi^2\varphi_{r_h}^2, \quad \lambda \equiv \frac{\mu^2}{v^2} \tag{40}$$

We see that the SM Higgs field corresponds to the radial Planck field in the spontaneously broken EW space. This is exactly given by the spacetime fluctuations in the radial direction of the potential, normalized by the Schwarzschild radius. The Higgs is not only the 'conduit' between the chirally left-handed and right-handed components of the fermions; it is also the bridge connecting the weak sector to the geometry. The

vacuum of the maximally symmetric stochastic spacetime is revealed as the Higgs vacuum post-symmetry breaking.

3.4. The Accretion Radius

Next, we consider the ground state potential at its minimum value in the neighborhood of the weak potential well at $r = r_a \equiv r_0 \sqrt{5+\sqrt{21}}$. We now take advantage of Equation (33) and the several helpful relationships corresponding to this point of minimum potential

$$\frac{r_0^2}{r_a^2}\left(\frac{1}{4}+\sqrt{\frac{r_a^2}{2r_0^2}-\frac{3}{16}}\right)=\frac{1}{4} \tag{41}$$

$$\frac{r_0}{r_a}+\frac{r_a}{2r_0}=\sqrt{\frac{7}{2}} \tag{42}$$

$$\frac{r_0}{r_a}\left(\frac{1}{4}-\sqrt{\frac{r_a^2}{2r_0^2}-\frac{3}{16}}\right)=-\frac{1}{2}\sqrt{\frac{3}{2}} \tag{43}$$

to obtain the ground state potential at the accretion radius, perturbed by the stochastic field at this point in space with $\delta\xi^r = \delta\xi^r_{r_a}$, as follows:

$$U(\tilde{r})_+ \to U(r_a,\delta\xi^r)_+ = m_\varphi^2\varphi_{r_h}^2\left\{\frac{9}{4}+\frac{3}{2}\sqrt{\frac{3}{2}}\frac{\delta\xi^r}{r_0}-\frac{3}{8}\frac{\delta\xi^{r\,2}}{r_0^2}-\frac{1}{4}\sqrt{\frac{3}{2}}\frac{\delta\xi^{r\,3}}{r_0^3}+\frac{1}{16}\frac{\delta\xi^{r\,4}}{r_0^4}\right\} \tag{44}$$

At the accretion radius, a variable transformation, $\tilde{z} \equiv \sqrt{\frac{3}{2}} - \frac{\delta\xi^r}{r_0} = \sqrt{\frac{3}{2}}(1-\sqrt{\frac{2}{3}}\frac{\delta\xi^r}{r_0})$, results in the potential as a function of $\tilde{z}$, as follows:

$$U(r_a,\tilde{z})_+ = const. + m_\varphi^2\varphi_{r_a}^2\left\{-\frac{15}{16}\tilde{z}^2+\frac{1}{16}\tilde{z}^4\right\} \tag{45}$$

The corresponding scalar doublet ϕ_a at r_a mirrors the scalar doublet at r_h, with h_a in place of h_s,

$$\phi_a = \frac{1}{\sqrt{3}}\begin{pmatrix}0\\-v\tilde{z}\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\-v+h_a\end{pmatrix}, \quad h_a \equiv v\sqrt{\frac{2}{3}}\frac{\delta\xi^r}{r_0} \tag{46}$$

The scalar doublet at r_a has a negative expectation value $\langle\phi_a\rangle_{\delta\xi^r} = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\-v\end{pmatrix}$. It is important to remember that the spacetime perturbation occurs at r_a and thus is independent of the perturbation at r_h: $\langle h_s h_a\rangle_{\delta\xi^r} = 0$.

The scalar field $\pm v$ degeneracy of the EW sector when mapped in space shows a clear difference between the two scalar fields ϕ_s and ϕ_a, corresponding to the positive and negative VEV solutions.

3.5. An Entire Family of Sombrero Hat Potentials

In formulating the EW $SU(2)_L$ ground state potential inside the Schwarzschild sphere, we picked a particular point in space, $r = r_h = \sqrt{5-\sqrt{21}}r_0$ and the corresponding transformation $\tilde{z} \equiv \sqrt{\frac{3}{2}} + \frac{\delta\xi^r}{r_0}$. In fact, any choice of $\tilde{z}$ given by

$$\tilde{z}_\pm(r) \equiv z_{0\pm}(1+\frac{1}{z_0}\frac{\delta\xi^r}{r_0}), \quad z_{0\pm} \equiv 2\frac{r_0}{r}\left(\frac{1}{4}\mp\sqrt{\frac{r^2}{2r_0^2}-\frac{3}{16}}\right) \tag{47}$$

results in the sombrero hat potential for $\varphi(\tilde{r})$, namely

$$U(r,\tilde{z}_\pm)_\pm = const. + m_\varphi^2 \varphi_{r_h}^2 \{-\frac{3}{4} Z_\pm(r)\tilde{z}_\pm^2 + \frac{1}{16}\tilde{z}_\pm^4\} \tag{48}$$

$$Z_\pm(r) \equiv 1 + \frac{r_0^2}{r^2}(\frac{1}{4} \pm \sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}}) \tag{49}$$

and the corresponding EW potential for $U_\pm^{SU(2)}$, namely

$$U_\pm^{SU(2)}(r) = const. - \mu^2\phi^\dagger\phi + f_\pm(r)\,\lambda\,(\phi^\dagger\phi)^2, \quad \mu^2 v^2 \equiv \frac{3}{2} z_{0\pm}^2 Z_\pm m_\varphi^2 \varphi_r^2, \quad \lambda \equiv \frac{\mu^2}{v^2} \tag{50}$$

where $f_\pm(r)$ defines the coupling relative to the SM coupling, as follows:

$$f_\pm(r) \equiv \frac{2}{3} \frac{(\frac{1}{4} \mp \sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}})^2}{\frac{r^2}{r_0^2} + (\frac{1}{4} \pm \sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}})} \tag{51}$$

The plot of the coupling coefficient $f_\pm(r)$ across r is shown in Figure 3.

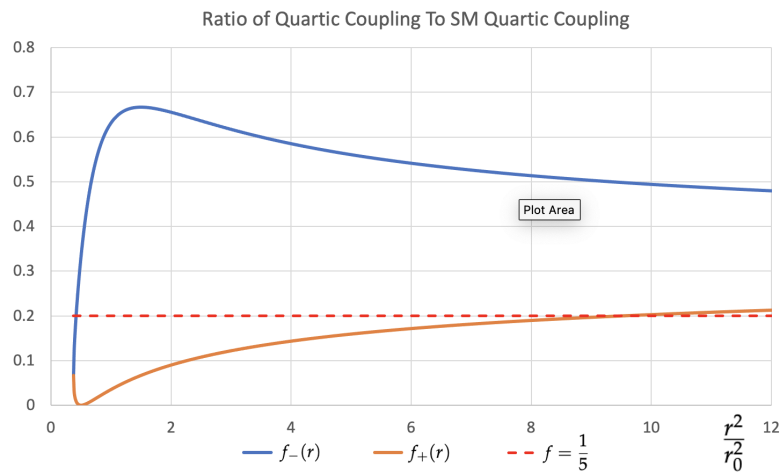

Figure 3. Comparison of quartic coupling coefficient to SM ($\frac{\lambda}{\lambda_{SM}}$) across $\frac{r^2}{r_0^2}$.

3.6. The Transition Point

To obtain the transition point, r_T, we require $\partial_r^2\varphi = -\frac{2\varphi}{\sigma^2\delta t}$ to equal $\partial_r^2\varphi$, calculated using Equation (31), such that

$$r_T^2 = \frac{7}{8} r_0^2$$

$$\partial_r^2\varphi = \partial_r(\frac{\varphi}{r}(\frac{1}{4} \pm \sqrt{\frac{2r^2}{\sigma^2\delta t} - \frac{3}{16}}))$$

$$= \frac{\varphi}{\sigma^2\delta t}(\,2\,(\,1 - \frac{1}{\sqrt{\frac{2r^2}{\sigma^2\delta t} - \frac{3}{16}}}\,) + \frac{\sigma^2\delta t}{r^2}\,(\,-\frac{3}{8} \pm \frac{1}{2\sqrt{\frac{2r^2}{\sigma^2\delta t} - \frac{3}{16}}}\,)\,) \tag{52}$$

$$= \frac{2\varphi}{\sigma^2\delta t}\,(\,1 - \frac{1}{\sqrt{\frac{2r^2}{\sigma^2\delta t} - \frac{3}{16}}}\,) + O(\delta t^0) \tag{53}$$

where we switched back to the δt notation so that we can easily see the order of δt in formulating the partial derivative. Setting Equation (52) directly to $-\frac{2\varphi}{\sigma^2 \delta t}$ and solving for r without taking into account the order of δt gives us an incorrect transition point just outside the horizon at $r = r_T^{incorrect} \approx \sqrt{1.83} r_0$. To correctly calculate the transition radius, we instead must require the second-order partial derivative to remain at the order of δt^{-1}. Thus, the correct formulation is given by Equation (53) and gives us $r = r_T = \sqrt{\frac{7}{8}} r_0$. The transition to the broken EW space can seamlessly occur just inside the Schwarzschild radius in the ground state.

There are two possible scenarios post-transition: a drop to the bottom of the Schwarzschild potential well along the ground state potential, or tunneling outside the Schwarzschild sphere and into the shallow well potential at the accretion radius, as shown in Figure 4.

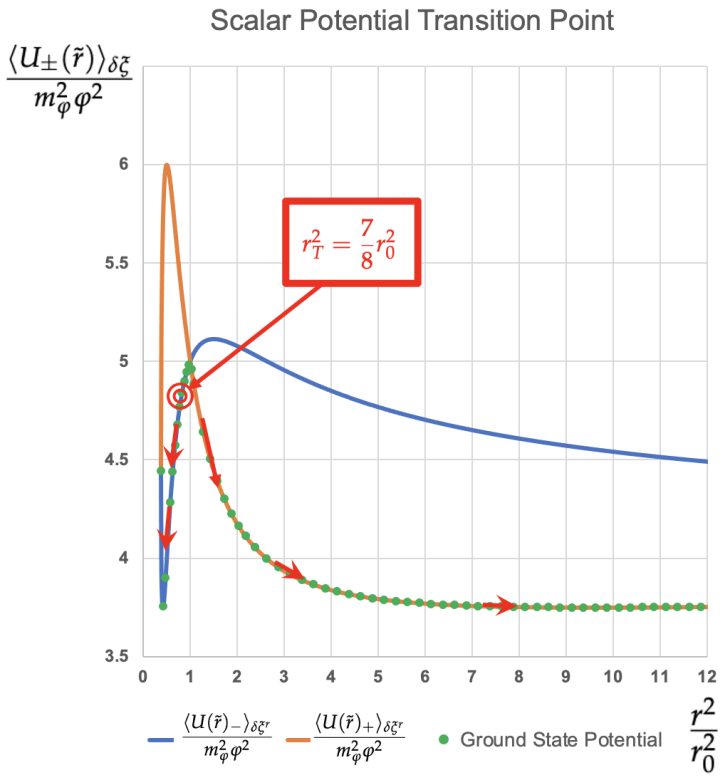

Figure 4. Transition point with drop to the minimum of the ground state potential inside the Schwarzschild black hole at r_h, or tunneling outwards and into the shallow well potential at r_a.

Finally, note that the requirement of $\partial_r \langle U(\tilde{r})\rangle_{\delta\xi} = 0$ is to order δt while $\langle U(\tilde{r})\rangle_{\delta\xi}$ is to order δt^2, and is therefore a 'loose' boundary condition. The boundary condition is satisfied exactly at the transition point, but it also holds for all values of r, across which the scalar field remains mostly flat.

3.7. The Nature of Symmetry Breaking and the Metastable Higgs Vacuum

Figure 5 shows the possible transitions at r_T between the maximally symmetric vacuum and the Higgs vacuum of the broken EW sector. A natural transition from the excited state of the maximally symmetric vacuum to the ground state of the Higgs vacuum is shown, along with the possible tunneling between the ground state of the Higgs vacuum and the ground state of the maximally symmetric vacuum. The metastable Higgs vacuum of the broken EW sector introduces the possibility of a bubble-nucleation-like process and begs the question of similarly statistically induced symmetry breaking [35,36]. Such

possibly probabilistic and spatially dependent dynamics need to be understood better and are left for future consideration.

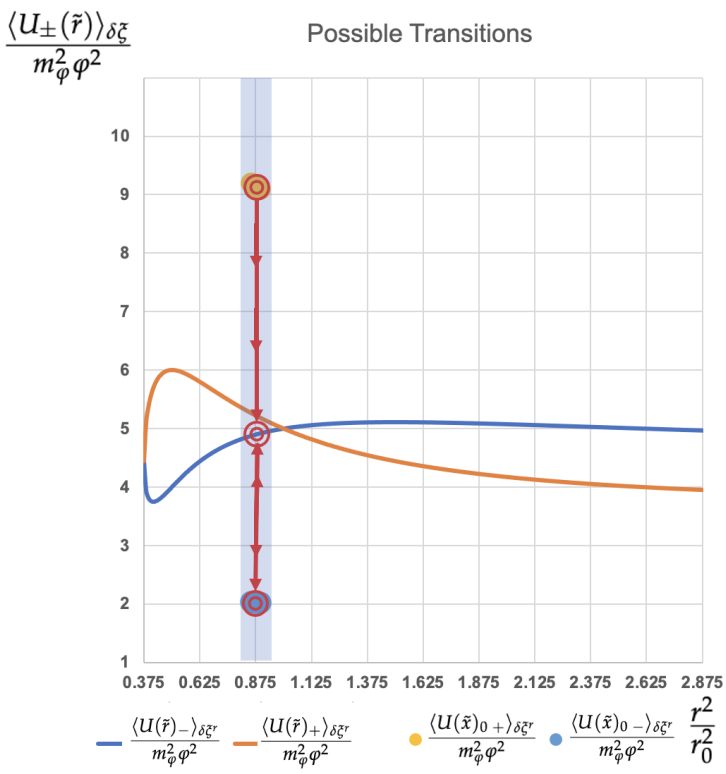

Figure 5. Possible transitions at r_T from the excited state of the maximally symmetric vacuum to the ground state of the Higgs vacuum. Possible tunneling between the ground state of the Higgs vacuum and the ground state of the maximally symmetric vacuum is also shown.

The rate of Higgs vacuum decay seeded by primordial black holes has been studied [37,38], and some of the literature suggests that it might be suppressed by quartic coupling and is consequently negligible.

The possibility of Reissner–Nordstom or Kerr black holes must also be considered as the charge and spin of primordial black holes play a critical role in Higgs vacuum stability [39–41]. It has been suggested that the rotation introduces curvature and, in an extreme case, might suppress Higgs vacuum decay, while moderate spin might have the opposite effect and enhance decay. Similarly, for very small or primordial black holes, a moderate charge can enhance Higgs vacuum decay via concentrated curvature and thermal effects, while an extreme charge suppresses decay by eliminating Hawking-induced stochastic fluctuations. Additional research must be performed to ascertain the effect of both spin and charge on the stochastically induced Schwarzschild black hole of the broken EW sector mapped in space considered here.

3.8. Conservation of ρ_{vev} and Restoration of SM λ

The factor of $\frac{1}{5}$ in the quartic coupling inside the Schwarzschild sphere and at the accretion disk emerges at the minimum of the potential in the revealed sombrero hat potential when mapped in space. To better understand the factor of $\frac{1}{5}$, consider what it tells us relative to the SM under the constraint that the observed value for the scalar VEV must match the well-established value of 246 GeV, as measured at the Large Hadron Collider (LHC).

In the SM, consider ρ_{vev}, defined by the minimum of the EW potential, and compare it to $\rho_{vev}^2 = \langle v^2 + h^2 \rangle_{\delta\xi}$:

$$U_{SM}^{SU(2)}(r >> r_0) = const. - \frac{1}{2}\mu^2\rho^2 + \frac{1}{4}\frac{\mu^2}{v^2}\rho^4 \tag{54}$$

$$\frac{\partial U_{SM}}{\partial \rho}|_{\rho=\rho_{vev}} = \mu^2\rho(-1 + \frac{\rho^2}{v^2}) = 0 \quad \Rightarrow \quad \rho_{vev} \approx v = 246\text{GeV} \tag{55}$$

$$\rho_{vev}^2 = \langle v^2 + h^2 \rangle_{\delta\xi} = v^2 \quad \Rightarrow v^2 >> \langle h^2 \rangle_{\delta\xi} \tag{56}$$

Thus the SM EW potential not only expects $v^2 >> \langle h^2 \rangle_{\delta\xi}$ but requires it.

Inside the Schwarzschild sphere, we instead have

$$U_{-}^{SU(2)}(r < r_0) = const. - \frac{1}{2}\mu^2\rho^2 + \frac{1}{5}\frac{1}{4}\frac{\mu^2}{v^2}\rho^4 \tag{57}$$

$$\frac{\partial U_{-}}{\partial \rho}|_{\rho=\rho_{vev}} = \mu^2\rho(-1 + \frac{\rho^2}{5v^2}) = 0 \quad \Rightarrow \quad \rho_{vev} = \sqrt{5}v = 246\text{GeV} \tag{58}$$

$$\rho_{vev}^2 = \langle v^2 + h^2 \rangle_{\delta\xi} = 5v^2 \quad \Rightarrow \langle h^2 \rangle_{\delta\xi} = 4v^2 >> v^2 \tag{59}$$

Inside the Schwarzschild black hole, it is the Higgs perturbation that dominates over v. We can rewrite the potential to reflect this as follows:

$$\rho_{vev}^2 = 5v^2 \quad \Rightarrow h_{vev} \approx \sqrt{5}v = 246\text{GeV} \tag{60}$$

$$U_{-}^{SU(2)}(r_h) = const. - \mu^2\phi^\dagger\phi + \lambda_{eff}(\phi^\dagger\phi)^2, \quad \lambda_{eff} \equiv \frac{\mu^2}{h_{vev}^2} \tag{61}$$

Higgs fields dominate ρ_{vev}, making the magnitude of the scalar field, v, effectively the perturbation in the Higgs vacuum.

We made a rough approximation for $h_{vev} \approx \sqrt{5}v = 246\text{GeV}$. The exact picture is a bit more complicated, with

$$\rho_{vev}^2 = h_{vev}^2 + v^2 = 5v^2 \quad \Rightarrow \quad v = \frac{1}{\sqrt{5}}246\text{GeV} \approx 110\text{GeV} \quad \Rightarrow \quad h_{vev} = 2v \approx 220\text{GeV} \tag{62}$$

This helps us understand where the factor of $\frac{1}{5}$ comes from: it is simply a statement of VEV preservation!

We also see that it is not v^2 which is preserved, but rather $\rho_{vev}^2 = v^2 + h_{vev}^2$. The correct potential for the Higgs vacuum that preserves the VEV regardless of the choice of mapping in EW space or geometric space naturally follows:

$$U^{SU(2)} = const. - \mu^2\phi^\dagger\phi + \lambda_\rho(\phi^\dagger\phi)^2, \quad \lambda_\rho \equiv \frac{\mu^2}{\rho_{vev}^2} \tag{63}$$

Have experiments at the LHC been effectively probing the mass $m_{\delta\rho}$ where $\rho \approx \rho_{vev} + \delta\rho$?

An interesting case to consider is $h_{vev} = v = \frac{1}{\sqrt{2}}246GeV \approx 173.95GeV$, where we cannot tell the difference between h and v and the potential is given by

$$U^{SU(2)} = const. - \mu^2\phi^\dagger\phi + \frac{\lambda}{2}(\phi^\dagger\phi)^2, \quad \lambda \equiv \frac{\mu^2}{v^2} \tag{64}$$

3.9. Quantum Mixture of ϕ_s and ϕ_a and Stochastic Entanglement

Both h_s and h_a are explicitly defined by the Planck fields and have well-defined two-point expectation values: $\langle h_s^2 \rangle_{\delta\xi} = \langle h_a^2 \rangle_{\delta\xi} = \frac{8}{3}v^2$. This suggests that a mixture of ϕ_s and ϕ_a

is required with the consequent Higgs field a linear combination of h_s and h_a in order to ensure the established VEV values $\rho^2_{vev} = 5v^2$ and $h^2_{vev} = 4v^2$.

While the topic of the quantum mixture of ϕ_s and ϕ_a deserves additional consideration, here, we simply note that $\phi_\pm = \frac{\sqrt{2}\pm 1}{2}\phi_s + \frac{\sqrt{2}\mp 1}{2}\phi_a$ results in $\rho = \pm v + h_\pm$, $h_\pm \equiv \frac{1}{\sqrt{2}}(h_s + h_a) \pm \frac{1}{2}(h_s - h_a)$, satisfying the above VEV requirements for ρ_{vev} and h_{vev}.

The mixed state also provides us with an estimate of the tunneling probability of the accretion radius by calculating the relative probability of finding a Higgs field at the accretion radius relative to inside the Schwarzschild sphere: $\frac{|\langle\phi_a\phi\rangle|^2}{|\langle\phi_s\phi\rangle|^2} \approx 1.13\%$.

The two scalar fields, one inside the Schwarzschild sphere and the other at the accretion radius, are effectively statistically entangled in order to ensure VEV preservation. How this relationship might relate to the information paradox [42] remains to be understood, and we leave it for future considerations.

4. Higgs Vacuum as an Over-Damped System in Equilibrium

We now consider the Higgs vacuum as a system in equilibrium corresponding to a highly damped state [14–17], consistent with the steep well potential inside the Schwarzschild sphere. We assume that the momentum change due to collisions is small and a soft-scattering approximation [18] can be applied to the Higgs fields, allowing for an application of the Smoluchovski equation for Higgs vacuum density evolution.

4.1. Ito Process in Spacetime, Fokker–Planck Equation, and Partition Function

We now switch to the review of the Fokker–Planck equation and corresponding partition function [26] for a general—albeit somewhat simplistic—multivariate Ito process. For **q**, a vector of hydrodynamic variables, a simple Langevin equation is given by

$$\delta q_i = F_i \delta t + \Delta_{ij}\sqrt{2\delta t}\eta_j \qquad \langle \delta q_i \rangle = 0 \qquad \langle \eta_i \eta_j \rangle = \delta_{ij} \tag{65}$$

The resulting Fokker–Planck equation is then [21,26,43,44]

$$\frac{\partial p(q,t)}{\partial t} = -\frac{\partial}{\partial q_i}(F_i\, p(q,t)) + \frac{\partial^2}{\partial q_i \partial q_j}(\Delta_{ik}\Delta_{jk}\, p(q,t)) \tag{66}$$

where summation over repeated indices is assumed, $i, j, k = 1 \ldots N_q$, with N_q being the number of hydrodynamic variables. The partition function follows [26]:

$$Z = (4\pi)^{-N_q/2} \int [\mathcal{D}q] \sqrt{\frac{\pi}{|\Delta^2|}} e^{-\frac{1}{4}\int d^4x \int d^4y (F(x)-\dot{q}(x))^T (\Delta^2(x,y))^{-1} (F(y)-\dot{q}(y))} \tag{67}$$

To shape the partition function to something we can use easily to obtain various point functions, we require the constitutive relations that define F and Δ. However, it is worth noting that it is much easier to work from here on in four-momentum space, allowing us to express point functions in terms of conjugates ω and **k** to t and **x**. We consider simple diffusion to illustrate this point.

For an example, consider a single diffusive process for a charge density n, and hence, a single constitutive relation [26]:

$$\dot{n}_x - D\nabla^2 n_x - \theta = 0 \tag{68}$$

To relate it to the partition function above, we identify $F_x = D\nabla^2 n_x$ and $\dot{q}_x = \dot{n}_x$. The fluctuations θ at some point in space x are treated as a divergence of a Gaussian noise in position and thus with a second moment given by $< \theta_x \theta_y >= -2TD\nabla^2\delta^4(x-y)$ [26]. This treatment of θ is consistent with the assumption that a kinetic random variable at some

point x is entirely defined by the same random variable at the same point x, an example of the traditional approach discussed above.

It is straightforward [26] to show that the Fourier transform of Δ^2_{xy} is given by $TD\mathbf{k}^2$ in order to obtain $(\Delta^2_{xy})^{-1} = \int \frac{d^4k}{(2\pi)^4} \frac{e^{-ik\cdot(x-y)}}{TD\mathbf{k}^2}$.

Equipped with Fourier transforms of F, $\dot{n}_x$, and $(\Delta^2_{xy})^{-1}$, we obtain the partition function [26]

$$Z = (4\pi)^{-N_q/2} \int [\mathcal{D}q] \sqrt{\frac{\pi}{|\Delta^2|}} e^{-\int d^4k n_k^2 \frac{(D\mathbf{k}^4+\omega^2)}{4TD\mathbf{k}^2}} \tag{69}$$

Further introducing a virtual source, completing the square, and integrating, we obtain the partition function and the two-point functions, all in momentum space, as follows [26]:

$$Z \propto e^{-\int d^4k \frac{TD\mathbf{k}^2}{(D^2\mathbf{k}^4+\omega^2)} J_k^2} \tag{70}$$

$$Z_0 \equiv Z|_{J=0} \tag{71}$$

$$\langle n_k \rangle = \frac{1}{Z_0} \left[\frac{\delta}{\delta J_k} Z \right]_{J=0} \tag{72}$$

$$\langle n_k n_{k'} \rangle = \frac{1}{Z_0} \left[\frac{\delta}{\delta J_k} \frac{\delta}{\delta J_{k'}} Z \right]_{J=0} \tag{73}$$

Thus, we finally obtain the expected density and the correlation

$$\langle n_k \rangle = 0, \quad G_{nn} \equiv \langle \delta n_k \delta n_k \rangle = \langle n_k n_k \rangle = \frac{2TD\mathbf{k}^2}{D^2\mathbf{k}^4 + \omega^2} \tag{74}$$

The detailed derivations and discussion regarding the path integral formalism are provided in [26], while here, we share the most relevant results as they pertain to the derivation of Higgs vacuum entropy and the correlation function.

4.2. Smoluchowski Equation for an Over-Damped System

As an introduction to the methodology necessary to obtain the dynamic characteristics of the Higgs vacuum density, entropy and correlation functions, we review a simple diffusive system in equilibrium characterized by an over-damped Langevin equation. We use this methodology to formulate the Langevin equation for the Higgs vacuum density and its solution under path integral formalism.

The Smoluchowski equation applied to an over-damped classical Brownian particle under some potential $V(\mathbf{r})$, mobility $\gamma \equiv \frac{2k_BT}{m\sigma_w^2}$, and a Gaussian noise w with $\langle d\mathbf{w}^2 \rangle_w = dt$ is given by [14–17],

$$d\mathbf{r} = -\frac{\sigma_w^2}{2k_BT} \nabla V(\mathbf{r}) dt + \sigma_w d\mathbf{w} \tag{75}$$

The path of such a particle describes a random walk, albeit highly constrained by the potential, with the resulting position and momentum carrying some degree of randomness.

The corresponding Fokker–Planck equation describing the evolution of this particle's probability density function, $p \equiv p(\vec{x}, t)$, is then given by

$$\partial_t p = \frac{\sigma_w^2}{2k_B T}(p\langle\nabla^2 V(\mathbf{r})\rangle_w + \langle\nabla V(\mathbf{r})\rangle_w \cdot \nabla p) + \nabla^2(\sigma_w^2 p) \tag{76}$$

For a Higgs field highly constrained by a steep well potential, we will take a similar approach and apply the corresponding Fokker–Planck equation to the Higgs field density. As will be shown, the resulting Fokker–Planck equation itself becomes a Langevin equation for Higgs field density.

4.3. Expectation Values for Functions of Both $\delta\xi$ and $\partial\delta\xi$

As we consider how the potential transforms during spontaneous symmetry breaking and in formulating the Fokker–Planck equation for the Higgs vacuum density, we are required to derive various derivatives of the scalar field and the potential. In accomplishing this, we must understand the appropriate order of δt for the derivatives as well as for expectation values of such derivatives. These are necessary steps required to properly consider the transformation of the scalar field and the potential as a result of spontaneous symmetry breaking.

A scalar field potential as a function of the stochastic spacetime, $U(\tilde{x})$ in Cartesian coordinates or $U(\tilde{r})$ in spherical coordinates, has no dependence on the kinetic fields. The same is not true in case of $\partial_r U(\tilde{r})$ or $\partial_r^2 U(\tilde{r})$, as we shall see.

To formulate the relationships between expectation values, let us work in a local frame at the Planck scale but in spherical coordinates with the Planck vector field in the radial direction, $\delta\xi^\mu \approx (0, \delta\xi^r, 0, 0)$. We can accomplish this generally, regardless of the order of δt chosen for the scalar field and thus the potential.

We can express the potential as a sum of terms corresponding to different powers of $\delta\xi^r$ and take the expectation value with respect to the same to obtain

$$U(\tilde{r}) = \sum_{n=0}^{\infty} U_n(r)(\delta\xi^r)^n \tag{77}$$

$$\langle U(\tilde{r})\rangle_{\delta\xi} = \sum_{n=0}^{\infty} U_n(r)\langle(\delta\xi^r)^n\rangle_{\delta\xi} \tag{78}$$

where $U_n(r)$ refers to the deterministic components of the potential as functions of r, corresponding to $\delta\xi^n$.

Using Equations (77) and (78), we can formulate $\partial_r U(\tilde{r})$, its expectation value, $\partial_{\delta\xi^r} U(\tilde{r})$, and $\partial_r\langle U(\tilde{r})\rangle_{\delta\xi}$, as follows:

$$\partial_r U(\tilde{r}) = \sum_{n=0}^{\infty} \partial_r U_n(r)(\delta\xi^r)^n + \sum_{n=0}^{\infty} nU_n(r)(\delta\xi^r)^{n-1}\partial_r\delta\xi^r \tag{79}$$

$$\partial_{\delta\xi^r} U(\tilde{r}) = \sum_{n=0}^{\infty} nU_n(r)(\delta\xi^r)^{n-1} \tag{80}$$

$$\langle\partial_r U(\tilde{r})\rangle_{\delta\xi} = \sum_{n=0}^{\infty} \partial_r U_n(r)\langle(\delta\xi^r)^n\rangle_{\delta\xi} + \sum_{n=0}^{\infty} nU_n(r)\langle(\delta\xi^r)^{n-1}\rangle_{\delta\xi}\,\partial_r\delta\xi^r \tag{81}$$

$$\partial_r\langle U(\tilde{r})\rangle_{\delta\xi} = \sum_{n=0}^{\infty} \partial_r U_n(r)\langle(\delta\xi^r)^n\rangle_{\delta\xi} \tag{82}$$

We see that ∂_r and the expectation value relative to $\delta\xi^r$ do not commute due to the kinetic stochastic term, $\partial_r\delta\xi^r$, which we take as independent of $\delta\xi^r$. However, note that

∂_r does commute with the expectation values taken with respect to both the stochastic spacetime fields and their kinetic counterparts, as follows:

$$\langle\langle\partial_r U(\tilde{r})\rangle_{\partial\delta\xi}\rangle_{\delta\xi} = \partial_r\langle U(\tilde{r})\rangle_{\delta\xi} = \partial_r\langle\langle U(\tilde{r})\rangle_{\partial\delta\xi}\rangle_{\delta\xi} \tag{83}$$

With the help of the above Equations (80)–(82), we obtain the intuitive—but not directly obvious—relationship

$$\langle\partial_r U(\tilde{r})\rangle_{\delta\xi} = \partial_r\langle U(\tilde{r})\rangle_{\delta\xi} + \langle\partial_{\delta\xi^r} U(\tilde{r})\rangle_{\delta\xi}\ \partial_r\delta\xi^r \tag{84}$$

A steep and narrow well potential will constrain the fields with $\partial^2\delta\xi^r \approx 0$ even at very high kinetic energy. In this case, following the same steps as above, we derive the expected value of the second-order partial derivative of the potential as follows:

$$\langle\partial_r^2 U(\tilde{r})\rangle_{\delta\xi} = \partial_r^2\langle U(\tilde{r})\rangle_{\delta\xi} + \langle\partial_{\delta\xi^r}^2 U(\tilde{r})\rangle_{\delta\xi}\ 2c_\xi\sigma_\xi^2\delta t + 2\partial_r\langle\partial_{\delta\xi^r} U(\tilde{r})\rangle_{\delta\xi}\ \partial_r\delta\xi^r \tag{85}$$

In Equation (85), we take advantage of Ito's Lemma in setting $(\partial_r\delta\xi^r)^2 = 2c_\xi\sigma_\xi^2\delta t$, consistent with the two-point function definitions we established in Equation (4).

These relationships will be useful as we formulate the transition of the scalar potential to reveal the broken EW symmetry and further consider the density and entropy of the Higgs vacuum.

4.4. Langevin and Fokker–Planck Equations for Higgs Fields

Fokker–Planck Equation (76) for a highly damped state requires the formulation of $\langle\partial^2 U(\tilde{r})\rangle_{\delta\xi^r}$, a rather tedious and perhaps confusing task. Luckily, we have already tackled the major questions involving taking derivatives of the potential, and we can simply use Equations (84) and (85) to write

$$\begin{aligned}\langle\partial_r^2 U(r,\delta\xi^r)\rangle_{\delta\xi^r} &= \langle\frac{1}{r^2}\partial_r(r^2\partial_r U(r,\delta\xi^r))\rangle_{\delta\xi^r} = \langle\frac{2}{r}\partial_r U(r,\delta\xi^r) + \partial_r^2 U(r,\delta\xi^r)\rangle_{\delta\xi^r}\\ &= \frac{2}{r}\partial_r\langle U(r,\delta\xi^r)\rangle_{\delta\xi^r} + \partial_r^2\langle U(r,\delta\xi^r)\rangle_{\delta\xi^r} + 8c_{\delta\xi}r_0^2\langle\partial_{\delta\xi^r}^2 U(r,\delta\xi^r)\rangle_{\delta\xi^r}\\ &\quad + 2\,\partial_r\delta\xi^r(\partial_r\langle\partial_{\delta\xi^r} U(r,\delta\xi^r)\rangle_{\delta\xi^r} + \frac{1}{r}\langle\partial_{\delta\xi^r} U(r,\delta\xi^r)\rangle_{\delta\xi^r})\end{aligned} \tag{86}$$

Substituting the relevant values in the Fokker–Planck equation for an over-damped system using Equation (76), we have

$$\partial_t p_\xi = -\frac{\sigma_\xi^2}{2k_BT}\ \{\langle\partial_r^2 U(r,\delta\xi^r)\rangle_{\delta\xi^r}\ p_\xi + \langle\partial_\mu U(r,\delta\xi^r)\rangle_{\delta\xi^r}\ \partial^\mu p_\xi\} + \sigma^2\ \partial^2 p_\xi \tag{87}$$

Assuming that we can approximate $\partial_r p_\xi|_{r_h} \approx 0$, consistent with an isotropic Higgs vacuum in equilibrium, we have

$$\partial_t p_\xi = \alpha_1\frac{\sigma^2}{2r_0^2}\ \frac{\mu^2 v^2}{k_BT}\ p_\xi + \sigma^2\ \partial_r^2 p_\xi + \alpha_2\frac{\sigma^2}{r_0^2}\ \frac{\mu^2 v^2}{k_BT}\ p_\xi\ \partial_r\delta\xi^r \tag{88}$$

While $m_\varphi^2\varphi^2 = \frac{16}{45}\mu^2 v^2$ technically holds for the neighborhood of r_h and r_a, we also assume that we can extend this relationship across r. With the above, the two dimensionless parameters α_1 and α_2 are given by

$$\alpha_1 \equiv \frac{r_0^2}{\mu^2 v^2}\{\partial_r^2\langle U(r,\delta\xi^r)\rangle_{\delta\xi^r} + 8c_\xi r_0^2\langle\partial_{\delta\xi}^2 U(r,\delta\xi^r)\rangle_{\delta\xi^r}\} \tag{89}$$

$$\alpha_2 \equiv \frac{r_0^2}{\mu^2 v^2}\{\frac{1}{r}\langle\partial_{\delta\xi} U(r,\delta\xi^r)\rangle_{\delta\xi^r} + \partial_r\langle\partial_{\delta\xi} U(r,\delta\xi^r)\rangle_{\delta\xi^r}\} \tag{90}$$

Note that the evolution equation for the Higgs density incorporates the stochastic effects of both the Higgs as well as the kinetic Higgs.

We can formulate the required terms to the appropriate order in δt with the help of Equations (29), (30), (84) and (86), in the range of r_T, as follows:

$$\partial_r^2\langle U_\pm(r,\delta\xi^r)\rangle_{\delta\xi^r} = 2\frac{m_\varphi^2\varphi^2}{r^2}(1 \pm 8\sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}}) + O(dt^1) \tag{91}$$

$$\langle\partial_{\delta\xi^r}^2 U_\pm(r,\delta\xi^r)\rangle_{\delta\xi^r} = 3\frac{m_\varphi^2\varphi^2}{r_0^2}(2 - \frac{r_0^2}{r^2}(\frac{1}{4} \pm \sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}})) + O(dt^1) \tag{92}$$

$$\langle\partial_{\delta\xi^r} U_-(r,\delta\xi^r)\rangle_{\delta\xi^r} = \frac{m_\varphi^2\varphi^2}{r}(6 - \frac{r_0^2}{r^2}) + O(dt^2) \tag{93}$$

$$\partial_r\langle\partial_{\delta\xi^r} U_\pm(r,\delta\xi^r)\rangle_{\delta\xi^r} = \mp 4\frac{m_\varphi^2\varphi^2}{r^2}\sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}} + O(dt^1) \tag{94}$$

Setting $c_\xi \sim \frac{1}{3\sigma_\xi^2\delta t} = \frac{1}{12r_0^2}$, we can approximate

$$\alpha_1 \approx \frac{m_\varphi^2\varphi^2}{\mu^2 v^2}\{4 + \frac{r_0^2}{r^2}(\frac{3}{2} \pm 14\sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}})\} \tag{95}$$

$$\alpha_2 \approx \frac{m_\varphi^2\varphi^2}{\mu^2 v^2}\,\frac{r_0^2}{r^2}\{6 - \frac{r_0^2}{r^2} \mp 4\sqrt{\frac{r^2}{2r_0^2} - \frac{3}{16}}\} \tag{96}$$

Figure 6 shows plots of $\alpha_{1\,-}$ and $\alpha_{2\,+}$ across r, corresponding to the two possible states based on the approximation $c_\xi \approx \frac{1}{12r_0^2}$. The remaining two parameters $\alpha_{1\,+}$ and $\alpha_{2\,-}$ are not shown but are somewhat larger and positive in magnitude. The negative range for $\alpha_{1\,-}$ will be significant when considering the entropy of the Higgs vacuum. It is important to keep in mind that we have approximated the value for the diffusion parameter c_ξ. Slightly smaller values of this parameter significantly widen the negative range for $\alpha_{1\,-}$ to also engulf the accretion region.

The probability density for the Higgs field itself, p_ξ, follows a Langevin equation. To transform the Fokker–Planck equation into a Langevin equation we can easily work with, again taking advantage of $\partial_r p_\xi \approx 0$, we make a variable substitution, providing us with the following Langevin equation for $ln\ p_\xi$:

$$q \equiv ln\ p_\xi \quad \Rightarrow \quad \partial_t q = \frac{\partial_t p_\xi}{p_\xi},\ \partial_r q = \frac{\partial_r p_\xi}{p_\xi} \approx 0,\ \partial_r^2 q = \frac{\partial_r^2 p_\xi}{p_\xi} - (\frac{\partial_r p_\xi}{p_\xi})^2 \approx \frac{\partial_r^2 p_\xi}{p_\xi} \tag{97}$$

$$\partial_t q = \alpha_1\frac{\sigma^2}{2r_0^2}\,\frac{\mu^2 v^2}{k_B T} + \sigma^2\,\partial_r^2 q + \alpha_2\frac{\sigma^2}{r_0^2}\,\frac{\mu^2 v^2}{k_B T}\,\partial_r\delta\xi^r \tag{98}$$

We solve for the Higgs vacuum density, correlation, and entropy via path integral formalism.

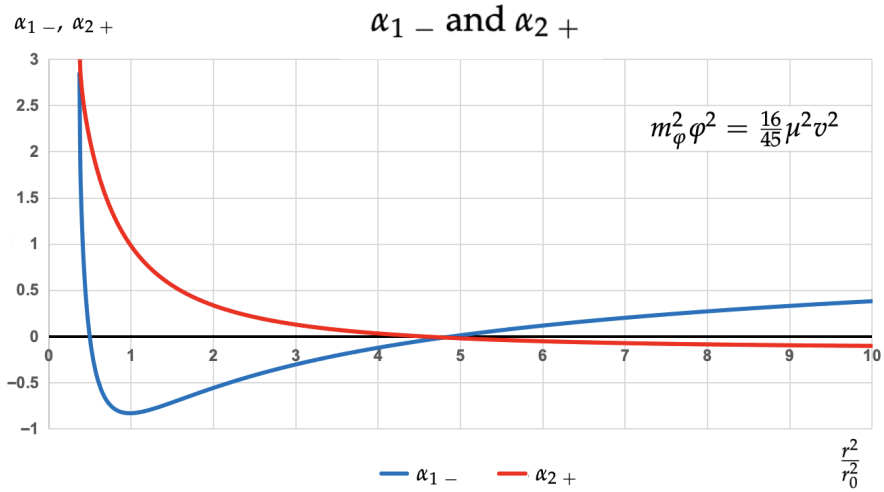

Figure 6. Parameters $\alpha_{1\,-}$ and $\alpha_{2\,+}$ plotted across r. The remaining two parameters $\alpha_{1\,+}$ and $\alpha_{2\,-}$ are not shown but are somewhat larger and positive in magnitude.

4.5. Higgs Vacuum Density, Correlation, and Entropy

For a Langevin equation of the form

$$\partial_t q_x = F_x + w_x \tag{99}$$

with $\langle w_x w_y \rangle_w \equiv 2\Delta^2_{x,y}\delta^4(x-y)$, the corresponding action derived via path integral formalism is given by

$$S = \int d^4x \int d^4y (F_x - \partial_t q_x)\frac{1}{\Delta^2_{xy}}(F_y - \partial_t q_x) \tag{100}$$

We can see by comparing Equation (98) to Equation (99) that in the case of the Higgs field, we obtain

$$F_0 \equiv \alpha_1 \frac{\sigma_\xi^2}{2r_0^2}\,\frac{\mu^2 v^2}{k_B T} \tag{101}$$

$$F_x \equiv F_0 + \sigma_\xi^2\,\partial_r^2 q \tag{102}$$

$$\Delta^2_{x,y} = \Delta^2 = 4c_\xi r_0^2\,\alpha_2^2\,\frac{\sigma_\xi^4}{2r_0^4}\,(\frac{\mu^2 v^2}{k_B T})^2 \tag{103}$$

Using Fourier transforms to work in the four-momentum space and adding a source term, we can write the partition function (keeping only the relevant terms) as follows:

$$Z \sim e^{\int d^4k \frac{(\Delta^2 J^2 + F_0 \sigma_\xi^2 \vec{k}^2 J)}{\sigma_\xi^4 \vec{k}^4 + \omega^2}} \tag{104}$$

Defining two more dimensionless parameters, a and b, for ease of formulation, we obtain

$$a \equiv \frac{\Delta^2}{\sigma_\xi^4\vec{k}^4 + \omega^2} = \alpha_2^2 \frac{\sigma_\xi^4}{r_0^2}(\frac{\mu^2 v^2}{k_B T})^2 \frac{4c_\xi}{\sigma_\xi^4\vec{k}^4 + \omega^2} \tag{105}$$

$$b \equiv \frac{F_0\,\sigma_\xi^2\vec{k}^2}{\sigma_\xi^4\vec{k}^4 + \omega^2} = \alpha_1 \frac{\sigma_\xi^2}{2r_0^2}(\frac{\mu^2 v^2}{k_B T})\frac{\sigma_\xi^2\vec{k}^2}{\sigma_\xi^4\vec{k}^4 + \omega^2} \tag{106}$$

the partition function and the resulting relevant point function are given by

$$Z[J] \sim e^{aJ^2+bJ} \tag{107}$$

$$\langle q^n \rangle_{\delta\xi} = \frac{\delta^n Z[J]}{\delta J^n}|_{J=0} \tag{108}$$

$$\langle lnp \rangle_{\delta\xi} = \langle q \rangle_{\delta\xi} \tag{109}$$

$$\langle p \rangle_{\delta\xi} = \langle e^q \rangle_{\delta\xi} = \sum_{n=0}^{n=\infty} \frac{1}{n!} \langle q^n \rangle_{\delta\xi} \tag{110}$$

$$\langle S \rangle_{\delta\xi} = -\langle p\, lnp \rangle_{\delta\xi} = -\sum_{n=0}^{n=\infty} \frac{1}{n!} \langle q^{n+1} \rangle_{\delta\xi} \tag{111}$$

We can consider the resulting density, correlation, and entropy according to which term dominates, a or b; however, it is not clear that this helps a great deal given the complexity of the point functions. Thus, to simplify the derivations, we can instead consider Higgs fields at very high energy corresponding to very fast oscillations inside the Schwarzschild sphere. Table 1 summarizes the various point functions along with the resulting entropy, Green's function provided by $G_{pp} \equiv \langle \delta p^2 \rangle_{\delta\xi} = \langle p^2 \rangle_{\delta\xi} - \langle p \rangle^2_{\delta\xi}$, and the Higgs vacuum density for the cases of $b >> a$, $a >> b$, and the most relevant case of high-energy Higgs fields where we keep only the lowest-order terms in $\frac{1}{\vec{k}^2}$, all in terms of a, b.

Table 1. Higgs vacuum density and entropy for the two Higgs vacuum states.

Vacuum State	$b >> a$	$a >> b$	High-Energy Fields
$\langle q^n \rangle_{\delta\xi}$	b^n	0 for n odd $1 \times 3 \times 5 \times ....(n-1)(2a)^{n/2}$ for n even	
$\langle p_k \rangle_{\delta\xi}$	e^b	e^a	$1+b$
G_{pp}	0	$e^{2a}(e^{2a}-1)$	$2a$
$\langle S_k \rangle_{\delta\xi}$	$-b\, e^b$	$-2ae^a$	$-b$

The density, Green's function, and the entropy can now be written out as

$$\langle p \rangle_{\delta\xi} = 1 + \alpha_1 \frac{\sigma_\xi^2}{2r_0^2} \left(\frac{\mu^2 v^2}{k_B T}\right) \frac{\sigma_\xi^2 \vec{k}^2}{\sigma_\xi^4 \vec{k}^4 + \omega^2} \tag{112}$$

$$G_{pp} = \frac{2}{3}\, \alpha_2^2 \frac{\sigma_\xi^4}{r_0^4} \left(\frac{\mu^2 v^2}{k_B T}\right)^2 \frac{1}{\sigma^4 \vec{k}^4 + \omega^2} \tag{113}$$

$$\langle S \rangle_{\delta\xi} = -\alpha_1 \frac{\sigma_\xi^2}{2r_0^2} \left(\frac{\mu^2 v^2}{k_B T}\right) \frac{\sigma_\xi^2 \vec{k}^2}{\sigma_\xi^4 \vec{k}^4 + \omega^2} \tag{114}$$

From Figure 6, we see that $\alpha_{1\,-}$, corresponding to the U_- solution for the potential, is negative starting at $r^2 = \frac{1}{2} r_0^2$, up to about halfway to r_a^2. Hence, inside the Schwarzschild sphere, the entropy of the ground state is positive with the exception of around the very core, close to the minimum possible value for r. Outside the sphere, the excited state has positive entropy but only about halfway to r_a^2, at which point the entropy of the excited state becomes negative. For the U_+ solution, the entropy is always negative as α_{1+} is always positive. Thus the entropy of the ground state outside the Schwarzschild sphere is negative.

This scenario is conditional on the value for the diffusion parameter c_ξ: slightly smaller values of c_ξ widen the range of positive entropy to include the accretion area. But perhaps even more importantly, if the effective Higgs field is the result of mixed quantum states, statistically entangled to ensure EW VEV preservation, entropy of the Higgs vacuum is not defined by the individual fields but rather by the mix.

Conditional on the chosen value for $c_{\tilde{\zeta}}$, the negative entropy contribution might come from Higgs fields in the accretion disk, which have a very small tunneling probability of around 1%. Thus, regardless of the value for $c_{\tilde{\zeta}}$, the positive entropy corresponding to ϕ_s dominates. This topic deserves a much deeper analysis, and we leave it for the future.

5. Additional Considerations and Applications

Finally, we will report here a few additional findings regarding particles in stochastic spacetime; however, most of the results presented here need additional research in order to arrive at some meaningful conclusion, and thus we leave it for the future. This being said, the results for the Dirac equation placed in a Planck field-infused vacuum suggest a new means for fermions to obtain mass and possibly impact Higgs field normalization.

The Dirac spinors placed in a stochastic spacetime, $\psi(\tilde{x})$, can be expanded out in powers of $\delta\xi^\mu$. Similarly, the covariant derivative must follow the algebra of stochastic spacetime and is given by $\tilde{\partial}_\mu = \partial_\mu - \partial_\mu \delta\xi^\alpha\, \partial_\alpha + \partial_\mu \delta\xi^\alpha\, \partial_\alpha \delta\xi^\beta\, \partial_\beta \ldots$ [11]. Keeping only the gauge-invariant terms, we find that the Dirac equation in stochastic spacetime reduces to the SM Dirac equation with an additional kinetic stochastic term, $\partial \cdot \delta\xi$, as follows:

$$\bar{\psi}(\tilde{x})(\gamma \cdot \tilde{\partial} + m)\psi(\tilde{x}) = \bar{\psi}(x)(\gamma \cdot \partial + m + m\, \partial \cdot \delta\xi)\psi(x) \tag{115}$$

Note that $\partial \cdot \delta\xi$ is a spin-0 scalar. The resulting coupling is very different from the standard Yukawa coupling as it involves coupling matter to field gradients rather than field amplitudes.

The expanded Dirac equation requires a renormalization of the Higgs field, as we now have for the one particle irreducible (1PI) two-point function of the kinetic Higgs a 1-loop $f\bar{f}$ contribution dominated by top quarks, as shown in Figure 7.

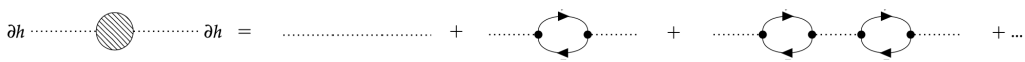

Figure 7. A 1PI two-point function of the kinetic Higgs dominated by the $t\bar{t}$ loop.

It remains to be seen how the quadratically divergent diagrams in the field normalization might impact the calculation of Higgs mass.

The mapping of the EW potential in space poses some interesting ideas regarding the imbalance of matter vs. anti-matter in the universe. The Schwarzschild black hole is a place where Higgs can become trapped, hiding matter. Regions where there might be a distinct signature of the give-and-take between h_{vev} and v might provide a clue.

Further probing the shape of the Higgs potential might test for the accretion disk effects and provide small deviations from SM predictions in Higgs partial widths. Additional consideration of the tri-Higgs coupling might give some clue of the possibility of the Higgs tunneling out of the black hole and into the accretion region where the sign of the coupling would flip.

6. Summary

We consider an $SU(2)$ EW scalar doublet ϕ placed in a maximally symmetric spacetime perturbed by stochastic Planck fields and in a standard quadratic potential. Constraining the scalar field to a zero VEV, we find the quadratic potential has a preferred direction parallel to the Planck fields.

In spacetime, a transformation to spherical coordinates reveals a statistically induced point mass, the result of Planck field correlations not visible in a maximally symmetric spacetime. Similarly, the transformation to spherical coordinates reveals a broken EW

sector and an entire family of possible ϕ^4 potentials with couplings, a function of the point in space.

Mapping the potential to EW sector corresponding to the two minima of the potential in space provides the SM sombrero hat potential, but with the quartic coupling instead reduced by a factor of $\frac{1}{5}$. The scalar field degeneracy in the EW sector corresponding to its positive and negative VEV can be mapped to the two potential minima in space, one inside the Schwarzschild sphere and inside a steep well potential, and another just outside in the accretion region and in a shallow well, respectively.

The factor of $\frac{1}{5}$ is a simple consequence of the conservation of EW VEV and the fact that the SM formulation of the EW potential does not account for situations where the perturbations in ϕ dominate. A more general formulation of the EW potential requiring the preservation of the EW VEV ρ_{vev} restores the SM quartic coupling.

Additional considerations of EW VEV invariance on the Higgs field VEV, h_{vev}, reveal the Higgs field as a statistically entangled linear combination of the two Higgs fields corresponding to the two minima in space, and allow for an estimate of the tunneling rate from the Schwarzschild sphere to the accretion region at roughly 1%.

While the Higgs vacuum corresponding to the deep well potential can be approximated to have positive entropy around the Schwarzschild horizon, there is a possibility of negative entropy in the accretion region. However, the Higgs vacuum as a mixture of the two states is dominated by the Higgs fields inside the Schwarzschild sphere corresponding to positive entropy.

The metastable Higgs vacuum of the EW sector introduces the possibility of a bubble-nucleation-like process with statistically induced symmetry breaking. Together with the possible impact charge and rotation might have on the Schwarzschild black hole, we leave the study of the stability of the Higgs vacuum in space to future considerations.

Funding: This research received no external funding.

Data Availability Statement: The data that support the findings of this study are openly available.

Acknowledgments: I am deeply grateful to the referees and editors of MDPI's Particles for their expertise and guidance in resolving several important questions and considerations. I would also like to thank Pavel Wiegmann and Petar Maksimović and his team; without their support, this research would not have been possible.

Conflicts of Interest: The author declares no conflicts of interest.

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